

Research

My research lies at the intersection of field robotics, machine learning/artificial intelligence, and ecology. I aim to develop autonomous systems, from novel learning-based algorithms to hardware platforms, capable of long-term monitoring and exploration, especially in highly dynamic environments, to better understand life on our planet.

Education

Massachusetts Institute of Technology and Woods Hole Oceanographic Institution 2019 – 2025

Ph.D. in Applied Ocean Science and Engineering & Electrical Engineering and Computer Science

Committee: Yogesh Girdhar (WHOI, advisor), Daniela Rus (MIT, collaborator), Sara Beery (MIT)

Thesis: “Tracking and Monitoring Marine Animals with Vision-guided Underwater Vehicles”

Massachusetts Institute of Technology 2016 – 2018

M.S. in Media Arts and Sciences, advised by Prof. Neri Oxman

Thesis: “On-site Autonomous Fabrication at Architectural Scales”

University of Pennsylvania 2008 – 2015

M.S.E. in Robotics, 2014-2015

B.S.E. in Computer Science, Minor in Mathematics, 2012

Academic Experience

University of Colorado, Boulder 2025 – Present

Postdoctoral Scholar in Computer Science and Environmental Data Science Innovation & Impact Lab (ESIL)

Advised by Prof. Esther Rolf

Massachusetts Institute of Technology and Woods Hole Oceanographic Institution 2019 – 2025

Advised by Dr. Yogesh Girdhar (WHOI), collaborated with Prof. Daniela Rus (MIT)

- Developed reinforcement learning algorithms for 6DOF control of autonomous underwater vehicles; first known zero-shot sim-to-real deployment on physical AUV [ICRA 2025].
- Designed autonomous systems for real-time visual tracking of marine animals using semi-supervised learning; first known to track a barracuda in the wild, 300 m distance [IJCV 2023].
- Formulated controls and experiments for novel agile robotic platforms including buoyancy-driven and bio-inspired locomotion for minimally invasive animal monitoring tasks [IEEE AUV 2022; ICRA 2023; ICRA 2024].
- Collaborated closely with biologists studying biodiversity, animal behavior, and robot-animal interactions in coral reef environments [ISER 2025].

University of Washington 2018 – 2019

Research Assistant in Aero&Astro and UW APL

Advised by Prof. Kristi Morgansen (UW AA) and Dr. Sarah Webster (UW APL), Graduate Researcher

- Designed automated tuners for Kalman filters used on UW Seaglidors [OCEANS 2019].

MIT Media Lab, Mediated Matter Group 2016 – 2018

Advised by Prof. Neri Oxman, Graduate Research Assistant

- Designed large-scale, mobile, autonomous robotic platform for 3D printing construction. Built on 4-DOF hydraulic cherry picker arm and a KUKA arm for fine-grain manipulation. Performed real-world demonstration of system by 3D-printing 14.6m diameter half-dome. Supported by *Alphabet's X (formerly Google [x])* [Sci. Rob. 2017, cover].
- Developed swarm-based robots that fabricate tubular fiberglass structures; created 20 robots that each autonomously built a 4m tall structure [CORO 2018; Sci. Rob. 2018].

University of Pennsylvania, GRASP Lab 2009 – 2015

Research Assistant in Electrical & Systems Engineering and Mechanical Engineering & Applied Mechanics

Advised by Prof. Daniel Lee and Prof. Mark Yim, Graduate Research Assistant

- Developed a provably optimal algorithm for any-angle path planning by applying a novel overestimate heuristic to the Fast Marching Method.
- Built software for user control of modular robots (CKBots) based on Robotics Bus Protocol.

Fellowships and Awards

3rd Place Best Poster Award – ICRA Robotics and Sustainability Workshop, 2024

WHOI Ocean Ventures Fund Recipient – Awarded \$10K to design modular, underwater, camera system

Nvidia Hardware Grant Recipient – Awarded an A6000 GPU for research purposes

NDSEG Fellow 2019 – 3 years full funding towards PhD

Link Ocean Engineering Fellowship 2019 – Declined for NDSEG (4 awarded annually)

UW GSFEI Top Scholar Program - Fellowship awarded to 1 student per department at Univ. of Wash.

KUKA Young Potential Best Paper Award – RobArch 2018 Conference paper

UPenn Rachleff Scholar – Highly-selective undergraduate research program

Publications and Presentations

* Co-first author

Journal Articles

- **L. Cai**, N. E. McGuire, R. Hanlon, T. A. Mooney. “Semi-supervised Visual Tracking of Marine Animals using Autonomous Underwater Vehicles.” *International Journal of Computer Vision (IJCV)*, 2023.
- M. Kayser*, **L. Cai***, S. Falcone, C. Bader, N. Inglessis, B. Darweesh, J. Costa, N. Oxman. “FIBERBOTS: An Autonomous Swarm-based Robotics System for Digital Fabrication of Fiber-based Composites.” *Springer Construction Robotics (CORO)*, Dec. 2018.
- M. Kayser*, **L. Cai***, S. Falcone, C. Bader, N. Inglessis, B. Darweesh, N. Oxman. “Design of a multi-agent, fiber composite digital fabrication system.” *Science Robotics*, Sept. 2018.
- S. J. Keating, J. C. Leland, **L. Cai**, and N. Oxman. “Toward site-specific and self-sufficient robotic fabrication on architectural scales.” *Science Robotics*, Apr. 2017. **Cover article.**

Peer-Reviewed Conference Papers

- **L. Cai**, Y. Jezequel, T. A. Mooney, Y. Girdhar. “Measuring and Minimizing Disturbance of Marine Animals to Underwater Vehicles.” *International Symposium on Experimental Robotics (ISER)*, 2025.
- **L. Cai***, K. Chang*, Y. Girdhar. “Learning to swim: reinforcement learning for 6DOF control of thruster-driven autonomous underwater vehicles.” *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- K. Macauley, **L. Cai**, P. Adamczyk, Y. Girdhar. “ReefGlider: A highly maneuverable vectored buoyancy engine based underwater robot.” *ICRA*, 2024.
- Y. Girdhar, N. McGuire, **L. Cai**, S. Jamieson, S. McCammon, B. Claus, J. E. San Soucie, J. E. Todd, T. A. Mooney. “CUREE: A Curious Underwater Robot for Ecosystem Exploration.” *ICRA*, 2023.
- J. Salazar, **L. Cai**, B. Cook, D. Rus. “Multi-Robot Visual Control of Autonomous Soft Robotic Fish.” *IEEE/OES Autonomous Underwater Vehicles Symposium (IEEE AUV)*, 2022.
- **L. Cai***, B. Boyacioğlu*, S. E. Webster, K. Morgansen. “Towards Auto-tuning of Kalman Filters for Underwater Gliders based on Consistency Metrics.” *MTS/IEEE OCEANS*, 2019.
- Y. Girdhar, **L. Cai**, S. Jamieson, N. McGuire, G. Flaspohler, S. Suman, B. Claus. “Enabling Co-Robotic Scientific Exploration of Unknown Environments over a Low Bandwidth Communication Channel.” *ICRA*, 2019.
- M. Kayser, **L. Cai**, C. Bader, S. Falcone, N. Inglessis, B. Darweesh, J. Costa, N. Oxman. “FIBERBOTS: Design and Digital Fabrication of Tubular Structures Using Robot Swarms.” *ROBARCH*, Aug. 2018. **KUKA Young Potential Best Paper Award.**

Peer-reviewed Workshop Proceedings

- **L. Cai**, Austin Greene, D. Yang, N. Aoki, S. Jarriel, J. Ha, N. Formel, T. A. Mooney, Y. Girdhar. "Species-level Detection and Tracking of Caribbean Coral Reef Fish." *IEEE International Conference on Computer Vision (ICCV) Workshop on Computer Vision for Ecology Proceedings*, 2025. (Archival, **oral spotlight**)

Workshop Presentations

- **L. Cai**, D. Yang, Y. Jezequel, T. A. Mooney, Y. Girdhar. "Quantifying Marine Animal Response to Surveys Using Underwater Vehicles." *ICRA Robotics and Sustainability Workshop*, 2024. **Best Poster 3rd Place Award**.
- **L. Cai**, D. Yang, S. Jamieson, Y. Girdhar. "Robot Goes Fish: Rapid, High-Resolution Biological Hotspot Mapping in Coral Reefs with Vision-Guided Autonomous Underwater Vehicles." *IEEE Computer Vision and Pattern Recognition (CVPR) CV4Animals Workshop*, 2023.
- **L. Cai**, N. E. McGuire, R. Hanlon, T. A. Mooney, Y. Girdhar. "Semi-supervised Visual Tracking of Marine Animals in the Wild." *CVPR CV4Animals Workshop*, 2022.
- **L. Cai**, R. Hanlon, Y. Girdhar. "Evaluation of Semi-supervised Visual Object Tracking Methods for Fully Autonomous In-situ, Tagless Tracking of Marine Animals." *CVPR CV4Animals Workshop*, 2021.
- "WARPAUV: A low-cost, vision-guided AUV for robotics research." Northeast Robotics Colloquium 2019.
- "Digital Construction Platform." Northeast Robotics Colloquium 2017.

Whitepapers/Preprints

- A. Papalia, C. Dawson, L. Anton, N. M. Bayomi, B. Champenois, J. H. Cho, **L. Cai**, J. DelPreto, K. Edwards, B. Githinji, C. Hickert, V. Jayawardana, M. Kramer, S. Raghavan, D. Russell, S. Salimi, J. Shi, S. Sudhakar, Y. Wang, S. Wang, L. Carlone, V. Kumar, D. Rus, J. Fernandez, C. Wu, G. Kantor, D. Young, H. Singh. "A Roadmap for Climate-Relevant Robotics Research." *Arxiv*, 2025.

Invited Talks

- **UW-Madison Guest Lecture** hosted by Prof. Wei Wang (Virtual, Sept. 2025).
- **SINTEF Ocean Invited Talk** hosted by Marios Xanthidis (Trondheim, Norway, Sept. 2025).

Selected Press

- **Wall Street Journal**, "Forget Humanoids. At MIT, Worms and Turtles Are Inspiring a New Generation of Robots." Isabelle Bousquette, May 2025.
- **PBS NOVA**, "Building Stuff: Change It! A Robot Made to Find Healthy Coral Reefs." Dec. 2024.
- **Nvidia Blog**, "Turning the Tide on Coral Reef Decline: CUREE Robot Dives Deep with Deep Learning." Scott Martin, Oct. 2023.
- **Mashable**, "Meet SoFi, MIT's soft robotic fish." Emmeth Smith, Jan. 2023.
- **Architect Magazine**, "3D Printing and Swarm Robotics Merge in Neri Oxman's Fiberbots." Wanda Lau, Oct. 2018.
- **Dezeen**, "Neri Oxman's swarm of Fiberbots autonomously build architectural structures." Natashah Hitti, Oct. 2018.
- **IEEE Spectrum**, "Watch this Robot 3D Print a Building Out of Spray Foam." Evan Ackerman and Alyssa Pagano, Apr. 2017.

Patents

- M. Kayser, L. Cai, S. Falcone, N. Oxman. "Methods and apparatus for tube fabrication." Patent application, Appl. No. US16/260,149.

Teaching and Mentoring

Supervised Master's or Undergraduate Projects

- Juan Salazar (MIT Aero, MEng.), Jadorian Paul (MIT EECS, undergrad), Braden Cook (MIT EECS, undergrad), Robert Williamson (MIT MechE, undergrad), Maaya Prasad (WHOI, EECS, undergrad), Charles Xu (WHOI, Physics, undergrad), Kevin Macauley (WHOI, MechE, undergrad), Kevin Chang (WHOI, CS, undergrad)

Massachusetts Institute of Technology, Electrical Engineering and Computer Science Dept.

- 6.S898, Deep Learning, grad-level, Fall 2023

University of Pennsylvania, Computer Science Dept.

- CIS520, Intro. to Machine Learning, grad-level, Fall 2015
- CIS521, Intro. to AI, grad-level, Fall 2012

Service and Outreach

ICRA'26 Associate Editor

ICRA'24 Lead Organizer for Workshop on Robotics for Understanding Natural Environments – Workshop to bring together roboticists and ecologists (<https://sites.google.com/view/icra-2024-rune-workshop>)

Reviewer for IEEE RA-L, IEEE ICRA, IEEE IROS, IEEE CVPR, IEEE ICCV, ISER, Springer Construction Robotics, IEEE BioRob, Elsevier Automation in Construction, Wiley Ecology and Evolution

Robotics: Science and Systems Conference (RSS) 2017 Volunteer

MIT RoboCon Organizer – Helped organize an MIT/Boston-area robotics conference (<http://robocon.mit.edu/>)

IBM Extreme Blue Technical Mentor – Mentored team of engineering undergraduates and an MBA student

FIRST Robotics Mentor – NCSSM Team 900 in Durham, NC

UPenn Summer Mentorship Program – Full-time instructor for 6-week, high-school summer technology course

FIRST Lego League Coach – FACTS Middle School, afterschool robotics team coach

Professional

MasterStreet (startup) *Software Engineer (New York, NY)* 2013 – 2014

- Built a search engine for continuing education courses with Ruby on Rails and ElasticSearch

IBM *Software Engineer (Durham, NC)* 2012 – 2013

- Developed mobile applications in Dojo for remote server management

Leadership and Extracurriculars

UPenn RoboCup Humanoid Soccer *Vision Team Lead* 2009 – 2012

UPenn FSAE Race Car *Electrical Team Lead* 2010 – 2012